Using a Decision Tree: Worked Example

The following example considers how two respective courses of action might be mapped out in practice.

New Store or Store Expansion?

Cara is a retail manager who has been asked by head office to respond to increased sales in her town by expanding store capacity. She is given the choice between using her building budget to expand her existing town centre store, or to invest in building a new out-of-town store. Her instinct is that building a new store will be far more effort than it's worth, but she thinks she should explore both options first, to test her gut feeling. She uses the decision tree technique to do so.

Building a new store would cost £400,000 more than the £1,000,000 cost of expanding the existing store, so these are the figures - £1,400,000 (new store) and £1,000,000 (store expansion) - that Cara sets out first, as the first two branches of her decision tree. She estimates that a new store would have a 0.5 chance of experiencing high sales and a 0.5 chance of experiencing low sales. By contrast, she estimates that an expanded existing store would have a 0.7 chance of experiencing high sales and a 0.3 chance of experiencing low sales. She adds two extra branches each to her original two options to reflect this.

Cara also calculates what the high and low turnover figures would mean in the first year in each scenario. For the new store, high first year turnover would mean £1,000,000, and low first year turnover £600,000. For store expansion, high first year turnover would be £400,000 and low first year turnover £100,000.

Next, Cara needs to multiply the probability of each turnover outcome occurring by the first year turnover itself, for both options.

New store with high first year turnover = £1,000,000 x 0.5 = £500,000.
New store with low first year turnover = £600,000 x 0.5 = £300,000.

Expanded store with high first year turnover = £400,000 x 0.7 = £280,000.
Expanded store with low first year turnover = £100,000 x 0.3 = £30,000.

Then, she needs to calculate what is known as the **total expected value**. This involves adding together your high first year forecast to your low first year forecast for each option. So, for the new store this is £800,000 and for store expansion this is £310,000. This step evens out the high and low turnover estimates by adding their respective probabilities together. It's essentially the equivalent of adding your 0.7 to your 0.3; after all, something in the range of these two figures will happen.

Finally, Cara needs to deduct her building cost from the total expected first year turnover value.

For the new store this is £800,000 - £1,400,000 = - £600,000.

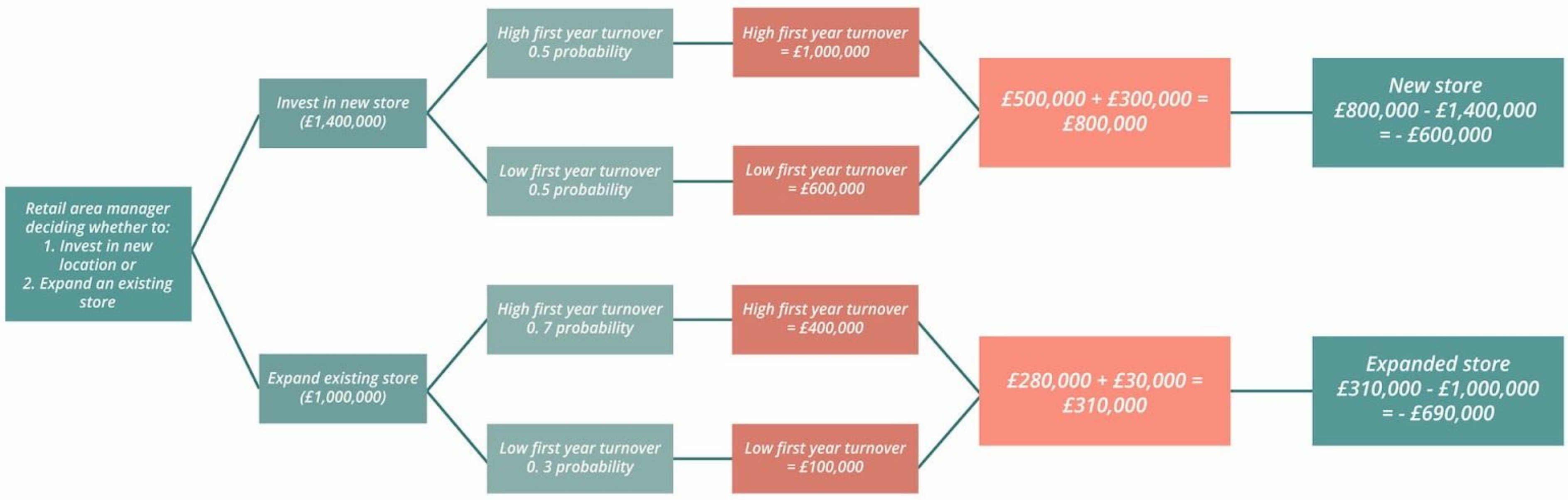
For the expanded store this is £310,000 - £1,000,000 = - £690,000.

Cara can then reflect on these two 'final' figures, which suggest that the cost of a new store isn't as relatively prohibitive as she first anticipated. Indeed, Cara is likely to be in less debt after the first year by building a new store.

The final decision she makes may depend on additional factors, but the use of this technique has expanded her viewpoint and given her serious food for thought.

Cara could continue adding branches to look at the probabilities of likely returns for subsequent years, too.

Cara's decision tree would look like this:



The final step, as described earlier, involves Cara weighing up the respective building costs against the first year's total expected value for each course of action. You'll recall that this step involved the following:

For the new store this is £800,000 - £1,400,000 = - £600,000.

For the expanded store this is £310,000 - £1,000,000 = - £690,000.